



CSE-4632

Digital Signal Processing Lab

Lab 1

Lab Group: 1

Thursday, 7th May, 2026

PART I

Task 1 — Continuous-Time Sinusoid & Signal Classification

Background: A continuous-time, continuous-valued (CTCV) sinusoid is defined as

$$x(t) = A \cos(2\pi f_0 t + \phi), \quad t \in \mathbb{R}.$$

Tasks:

- Generate and plot $x(t) = 3 \cos(2\pi \cdot 5 t + \pi/4)$ over $t \in [0, 1]$ seconds using a very fine time vector (`t = 0, 1, 1e-4`).
- Identify the **amplitude**, **frequency** (f_0), **angular frequency** (ω_0), and **phase** (ϕ) of $x(t)$.

Task 2 — Sampling a Sinusoid & the Nyquist Criterion

Background: Sampling converts a continuous-time signal to a discrete-time signal by recording its value at equally-spaced instants $t = nT_s$, where $T_s = 1/f_s$ is the sampling period.

Tasks:

- Start from the CTCV signal $x(t) = \cos(2\pi \cdot 10 t)$. Sample it at $f_s = 100$ Hz to obtain the discrete-time sequence $x[n] = x(nT_s)$. Plot $x[n]$ using `stem`.
- Repeat the sampling at $f_s = 15$ Hz (below Nyquist). Plot the result on the same figure and describe the aliasing effect you observe.
- Compute the *minimum* sampling frequency required to avoid aliasing for $x(t)$.

Task 3 — Discrete-Time Continuous-Valued (DTCV) Signal

Background: A DTCV signal is defined only at integer sample indices n but can take any real amplitude value.

Tasks:

- a) Generate the DTCV sequence

$$x[n] = 0.8^n \cos\left(\frac{\pi}{6}n\right), \quad n = 0, 1, 2, \dots, 30.$$

- b) Plot $x[n]$ with `stem` and label the axes correctly (use n on the horizontal axis, *not* time t).
- c) Find, analytically and verify in code, the **range** (maximum and minimum value) of $x[n]$ over $n \in [0, 30]$.

Task 4 — Periodicity of Discrete-Time Sinusoids

Background: A discrete-time sinusoid $x[n] = \cos(\omega_0 n)$ is periodic with period N if and only if $\omega_0/(2\pi)$ is a *rational* number, i.e. $\omega_0 = 2\pi k/N$ for integers k and N .

Tasks:

- a) For each signal below, determine analytically whether it is periodic; if so, find the fundamental period N :

i. $x_1[n] = \cos\left(\frac{\pi}{4}n\right)$

ii. $x_2[n] = \cos\left(\frac{3\pi}{7}n\right)$

iii. $x_3[n] = \cos(\sqrt{2}\pi n)$

iv. $x_4[n] = \cos\left(\frac{2\pi}{5}n\right) + \cos\left(\frac{2\pi}{3}n\right)$

- b) Verify your answers by generating upto $3N$ samples of each periodic signal and confirming $x[n + N] = x[n]$ numerically.

Task 5 — Frequency Identification from a Sampled Signal

Background: Given a sampled signal and knowledge of the sampling rate, the physical frequency of each sinusoidal component can be recovered from its *normalised digital frequency* $\hat{\omega} = \omega_0/f_s$.

Tasks:

- a) You are given the following sampled sequence (sampling rate $f_s = 1000$ Hz):

$$x[n] = 2 \cos(0.2\pi n)$$

Identify the **normalised digital frequencies**, **physical frequencies** (in Hz), and **phase**.

Task 6 — Phase Shift and Time Delay

Background: A phase shift ϕ in a sinusoid $A \cos(2\pi f_0 t + \phi)$ corresponds to a time delay $t_d = -\phi/(2\pi f_0)$.

Tasks:

- a) Generate the following three signals at $f_s = 500$ Hz over $t \in [0, 0.2]$ s:

$$x_1(t) = \cos(2\pi \cdot 10 t), \quad x_2(t) = \cos\left(2\pi \cdot 10 t + \frac{\pi}{2}\right), \quad x_3(t) = \cos\left(2\pi \cdot 10 t - \frac{\pi}{3}\right).$$

- b) Plot all three on the same figure. From the graph, estimate the time delay of x_2 and x_3 relative to x_1 .
- c) Verify your estimates using the formula $t_d = -\phi/(2\pi f_0)$.
- d) Identify which signal leads and which lags x_1 .